

- 4 The photograph shows a flightless grasshopper, *Brachaspis robustus*.



(Source: © Davide Bonora/Shutterstock)

This endangered species is only found in one small part of New Zealand.

These grasshoppers are approximately 4 cm long.

The grasshoppers are most active between November and March.

The grasshoppers are camouflaged so they can only be seen when they move.

A student decided to compare the population of grasshoppers living on an unused gravel road and a natural gravel area.

The student formed the following hypothesis:

The population of *B.robustus* in the natural gravel area will be larger than the population on an unused gravel road.

Plan an investigation to find evidence to support or reject this hypothesis.

- (a) Describe preliminary practical work that you might undertake to find a suitable method for observing these grasshoppers to provide quantitative results.

(3)

[illegible]

(b) Devise a detailed method, including how you would control and monitor important variables.

(8)



(c) Describe how your results should be recorded, presented and analysed in order to draw conclusions from your investigation.

(3)

[illegible]

(Total for Question 4 = 17 marks)

TOTAL FOR PAPER = 50 MARKS